Experiment 6.1.1

Algorithm: Incremented Date

Step 1: Start

Step 2: Input day, month, and year

Step 3:
If year ≤ 0 **OR** month < 1 **OR** month > 12,
    print **"Invalid Date"** and Stop

Step 4: Determine the maximum number of days in the given month

- For January, March, May, July, August, October, December → maxDays = 31
- For April, June, September, November → maxDays = 30
- For February:
    If leap year → maxDays = 29
    Else → maxDays = 28

Step 5:
If day < 1 **OR** day > maxDays,
    print **"Invalid Date"** and Stop

Step 6:
If day < maxDays,
    set day = day + 1

Step 7:

Else

set day = 1

set month = month + 1

Step 8:

If month > 12,

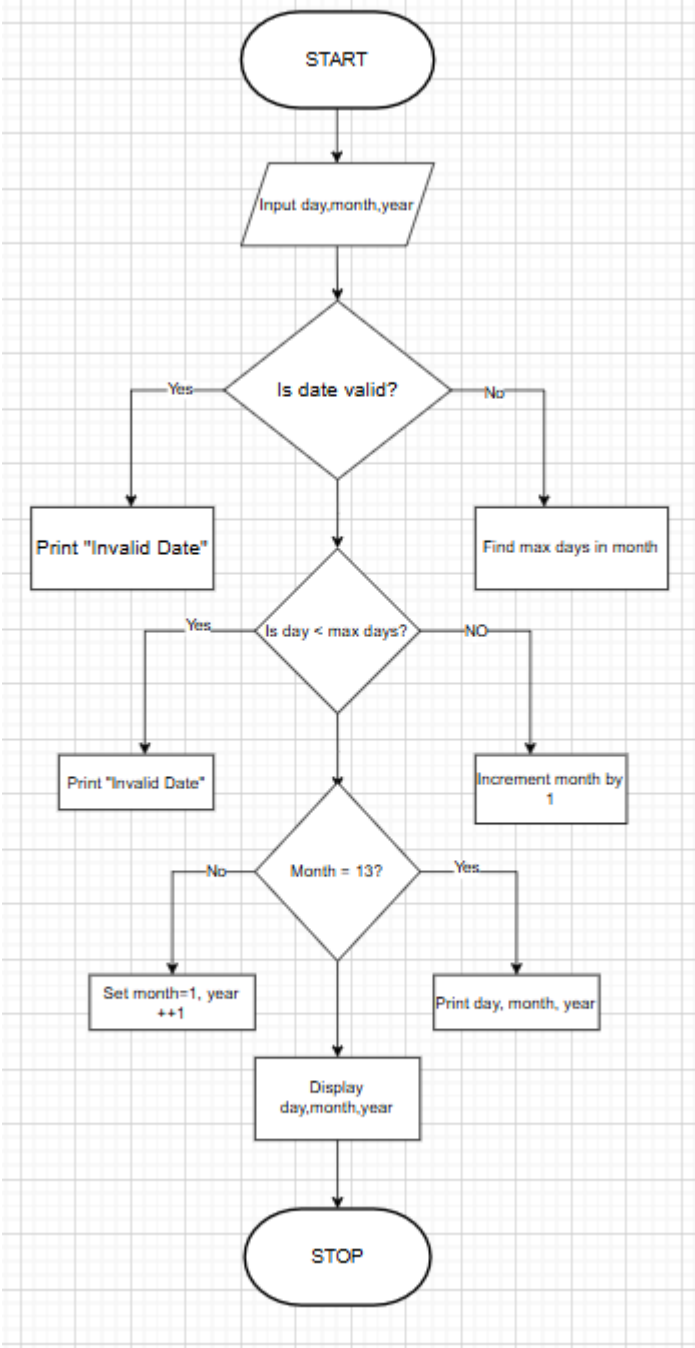
set month = 1

set year = year + 1

Step 9: Display the incremented day, month, and year

Step 10: Stop

Flowchart:


START
Input day,month,year
Is date valid?
Yes
No
Print "Invalid Date"
Find max days in month
Is day < max days?
Yes
NO
Print "Invalid Date"
Increment month by 1
Month = 13?
No
Yes
Set month=1, year ++1
Print day, month, year
Display day,month,year
STOP

Code:

CODETANTRA Home

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6.1.1. Incremented Date

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

1. **Valid Month**: 1 to 12
2. **Valid Day**: Depends on the month and year
   - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
   - April (4), June (6), September (9), November (11): 1 to 30 days
   - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
3. **Valid Year**: Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:

```
<DD>-<MM>-<YYYY>
```

- If the date is invalid, print: "Invalid Date".

Sample Test Cases +

Explorer

nextDate.py

Submit

```python
d = int(input())
m = int(input())
Y = int(input())

flag = True

if Y < 1 or m < 1 or m > 12:
    flag = False

    if m in (1,3,5,7,8,10,12):
        maxdays = 31
    elif m in (4,6,9,11):
        maxdays = 30
    else:
        if(Y % 4 == 0):
            maxdays = 29
        else:
            maxdays = 28

    if d > maxdays or d <1:
        flag = False

    d = d+1
    if d > maxdays:
        d=1
        m=m+1

    if m > 12:
        m=1
        Y=Y+1

    if flag:
        print(f"{d:02d}-{m:02d}-{Y}")

    else:
        print("Invalid Date")
```

Debugger

Terminal Test cases

< Prev Reset Submit Next >